

ABU DHABI INTL, United Arab Emirates

WMO: 412170

Lat: 24.433N

Lon: 54.651E

Elev: 88

StdP: 14.65

Time Zone: 4.00 (E04)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	53.6	55.7	30.6	24.8	88.0	35.4	30.3	86.4	21.6	69.7	19.6	70.9	4.9	120	0.456

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	21.8	113.2	73.5	111.1	73.9	108.8	74.1	87.0	95.5	86.0	94.7	85.1	94.2	9.7	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
84.6	182.6	92.5	84.1	179.2	92.3	82.7	170.9	91.7	52.2	95.3	50.9	94.6	49.8	94.4	91.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.9	18.7	17.1	DB	49.6	117.6	2.4	1.6	47.9	118.8	46.5	119.7	45.1	120.6	43.4	121.8	(4)
(5)			WB	44.5	88.9	3.2	1.2	42.2	89.8	40.3	90.5	38.5	91.2	36.2	92.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	83.7	66.9	69.8	75.2	82.9	90.3	93.9	97.4	98.6	93.6	86.7	77.8	70.2
	DBStd	11.72	3.82	5.32	6.01	5.78	4.47	3.77	4.25	3.15	3.06	3.30	3.76	3.82
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	35	21	9	0	0	0	0	0	0	0	0	0	5
	CDD50	12295	525	556	781	988	1250	1317	1468	1505	1308	1137	834	626
	CDD65	6856	81	145	317	538	785	867	1003	1040	858	672	384	166
	CDH74	96003	436	1031	2790	6431	11602	13657	16556	17035	13106	8845	3595	919
	CDH80	62098	75	374	1243	3503	7520	9399	12098	12571	8823	4969	1354	169
Wind	WSAvg	8.4	7.6	8.9	9.3	8.9	8.8	8.9	9.0	8.9	8.2	7.5	7.2	7.2
Precipitation	PrecAvg	2.2	0.4	0.4	0.5	0.2	0.1	0.0	0.0	0.0	0.0	0.0	0.1	0.4
	PrecMax	6.5	2.0	3.6	2.3	2.2	1.4	0.9	0.8	1.2	0.1	0.2	1.0	2.5
	PrecMin	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	1.8	0.6	0.7	0.7	0.5	0.3	0.2	0.2	0.2	0.0	0.0	0.2	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	84.9	93.4	100.7	107.3	111.7	113.8	116.0	116.0	111.3	104.1	95.3	87.9
		MCWB	63.8	63.5	65.3	68.4	69.5	71.5	73.9	73.8	72.1	70.9	67.1	65.8
	2%	DB	82.3	89.2	95.4	103.6	109.0	110.6	113.1	113.1	107.8	100.7	91.9	84.2
		MCWB	63.2	63.7	65.0	67.4	69.7	72.6	74.5	74.0	73.3	70.8	68.8	66.0
	5%	DB	78.9	84.5	91.4	99.6	105.9	107.5	110.6	110.9	105.6	98.6	89.7	81.5
		MCWB	62.4	62.8	64.7	67.2	70.4	73.1	75.3	74.9	73.9	71.5	69.6	66.8
	10%	DB	75.5	80.5	87.4	95.4	102.4	104.3	107.3	107.8	102.5	96.3	87.4	79.0
		MCWB	62.5	62.5	64.7	67.3	71.0	74.8	76.8	76.0	75.7	73.0	70.1	66.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	69.1	70.2	73.3	76.6	82.0	86.2	88.6	88.2	86.9	83.7	77.8	73.6
		MCDB	76.6	79.9	83.6	87.4	92.1	94.9	96.6	96.9	94.1	90.7	86.0	79.6
	2%	WB	67.2	68.3	71.4	74.8	80.1	84.6	87.0	87.2	85.4	82.0	75.6	71.2
		MCDB	74.9	76.7	81.0	86.1	90.5	93.6	95.4	96.0	93.5	89.7	84.6	78.0
	5%	WB	65.7	66.9	70.0	73.1	78.8	83.5	85.9	86.2	84.4	80.7	74.3	69.7
		MCDB	73.8	75.7	79.7	85.4	89.7	93.0	94.9	95.0	92.8	89.2	83.1	76.8
	10%	WB	64.4	65.4	68.7	71.7	77.3	82.2	84.7	85.2	83.5	79.3	72.8	68.3
		MCDB	72.5	75.0	78.5	84.1	89.7	92.5	94.5	94.5	91.9	88.1	81.9	76.0
Mean Daily Temperature Range	5% DB	MDBR	17.6	18.6	20.1	23.0	24.9	23.6	20.9	21.8	21.8	21.4	18.9	17.6
		MCDBR	22.2	24.9	26.5	29.0	29.0	27.4	24.1	24.8	24.5	23.7	22.0	20.8
		MCWBR	10.0	10.2	10.4	11.4	11.8	12.1	12.4	12.6	12.8	11.5	9.5	9.5
	5% WB	MCDBR	18.4	19.1	20.1	23.5	25.9	23.4	20.5	21.3	21.7	21.0	19.3	18.3
		MCWBR	10.1	9.9	9.7	10.6	11.2	10.8	10.7	10.9	10.7	10.2	9.2	9.6
Clear-Sky Solar Irradiance	taub	0.449	0.502	0.569	0.605	0.592	0.634	0.727	0.700	0.598	0.506	0.458	0.439	
	taud	2.017	1.869	1.720	1.673	1.688	1.643	1.529	1.578	1.728	1.905	2.030	2.050	
	Ebn at Noon	251	246	237	232	234	223	203	207	226	241	246	248	
	Edh at Noon	49	61	75	80	78	81	91	87	73	58	48	46	
All-Sky Solar Radiation	RadAvg	1339	1580	1858	2118	2346	2385	2234	2164	2018	1766	1436	1259	
	RadStd	73	122	122	123	63	49	73	45	40	30	35	73	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.95	+1.21	N/A	N/A	N/A	N/A	N/A	+340	+335
(47) Regional (9 neighbors)		+0.76	+1.08	N/A	N/A	N/A	N/A	N/A	+270	+268

Nomenclature: See separate page